WORLD SUSTAINABLE BUILDING 2014
BARCELONA CONFERENCE

Sustainable Building: RESULTS

Are we moving as quickly as we should?

It's up to us!

TRANFORMING REALITY CONCLUSIONS

This volume gathers the conclusions from the oral sessions from the Conference area "Transforming Reality", presented at the World SB14 Barcelona Conference, which took place in Barcelona on the 28th, 29th and 30th October 2014. This area presented sessions sponsored by different companies and entities, international stakeholders of sustainable building.

The Conference was organised by GBCe (Green Building Council España), co-promoted by iiSBE, UNEP-SBCI, CIB and FIDIC, and counted on the participation of World GBC*.

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***iiSBE:** International Initiative for a Sustainable Built Environment
UNEP-SBCI: United Nations Environment Programme - Sustainable Buildings and Climate Initiative
CIB: Conseil International de Batîment
FIDIC: International Federation of Consulting Engineers
World GBC: World Green Building Council

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Session 2:

Buildings: World's Closing Window of Opportunity; What could be the role of Rating Tools?

Chairperson:

Sauer, Bruno

Technical Director, GBCe

Speakers:

Kapoor, Prashant
Principal Industry Specialist-Green Buildings, International Finance Corporation (IFC). Washington.USA.
Oussidhoum, Firdaous
Director of International and External Relations at Africa Union Of Architects. Morocco.
Platt, Roger
President of US Green Building Council. USA.
Madew, Romilly

Chair of WorldGBC and Australia GBC. Australia. South Africa.

Conclusions:

The introduction of the Assessment Tool EDGE in the worldwide market of rating tools shows that there is a real need for bottom up simplified tools. EDGE focuses mainly on three aspects of sustainable building: energy, water and materials. It is an easy-to-use approach of a rating tool that aims to have a larger impact on policy and massive application. The fact that an international established and recognized tool like LEED is recently focusing more on social indicators shows that we are moving towards a worldwide tendency of balancing the environmental, social and economic criteria. There are 5 key priorities for social indicators: employment, education, equality, health & safety, and community management. However, despite of these insights and the efforts of EDGE to be adapted to countries in development, there is still a huge gap with the reality and needs of the African continent. Africa will have the largest growth of population in cities the coming decades and they are asking openly not being imposed external rating tools as guidelines for their development. They have a need of reinforcing their own models focusing on culture as a design value and community management as the fundamental base for sustainable development.

Session 3:

Which are the strategies and actions of the Government of Spain to achieve energy efficient building renovation and fight climate change?

Chairman:

Isabel León

Architect, Executive Committee of GBCe

Speakers:

Martín Ramiro, Francisco Javier
Subdirector General de Vivienda y Suelo, Dirección General de Arquitectura, Vivienda y Suelo, Ministerio de Fomento. Spain.
Gonzalez, Eduardo
Subdirector General de Coordinación de Acciones frente al Cambio Climático, Ministerio de Agricultura, Alimentación y Medio Ambiente. Spain.
Ruiz Castellanos, Jesús

Dpto. Doméstico y Edificios, IDAE. , Ministerio de Industria, Energía y Turismo. Spain.

Conclusions:

Creemos que España ha dado importantes pasos hacia delante en relación a las leyes y normativas que ha aprobado para renovar el sector de la edificación y reducir sus impactos.
Como la sesión tenía traducción simultánea, la sesión se desarrolló en español.
Los tres ponentes, representantes de los tres ministerios intervinieron según programa.
En primer lugar yo hice una breve presentación de los objetivos que perseguía esta sesión, las estrategias a largo plazo y las acciones del Gobierno de España para alcanzar los objetivos propuestos en el artículo 4 de la Directiva europea acerca de la renovación de edificios.
A continuación intervino Javier Martín, subdirector general de vivienda y suelo. Expuso brevemente la situación de nuestro parque inmobiliario y las acciones que estaba llevando a cabo el Ministerio de Fomento para paliar la situación, la Ley de Rehabilitación, Regeneración y Renovación Urbana aprobada el pasado año y La Estrategia A Largo Plazo para la Rehabilitación Energética en el Sector de la Edificación.
Seguidamente tomo la palabra, Eduardo González Subdirector de la Oficina de Cambio Climático, del Ministerio de Agricultura, Alimentación y Medio Ambiente .
Realizó una breve exposición acerca de la situación española en relación a emisiones de gases efecto invernadero y el cumplimiento del objetivo 20-20-20 y se detuvo en presentar el Proyecto Clima para impulsar la economía baja en carbono.
Finalmente intervino Jesús Ruiz Castellano, técnico del departamento de edificios. en representación del IDAE, Instituto para la Diversificación y Ahorro de la Energía, que explicó ampliamente el Programa PAREER, programa de ayuda a proyectos integrales de ahorro y eficiencia energética en edificios de viviendas.
También presentó brevemente el Fondo JESSICA, fondo de inversión para financiar proyectos de eficiencia energética y energías renovables
A continuación se produjo el debate que, como los ponentes se habían demorado algo más de

lo previsto, duro solo 20 minutos.
Las cuestiones sobre las que se interesaron fueron:
En primer lugar, tras felicitar a España por todas las acciones que estaba promoviendo, requirieron la necesaria coordinación entre Ministerios para alcanzar la máxima eficiencia en la aplicación de los planes que estaban desarrollando cada uno de ellos.
Hubo un debate donde se propuso la ventanilla única para la gestión de la rehabilitación. Los tres ponentes explicaron las dificultades que ello conlleva.
Otro tema de largo debate se suscitó cuando preguntaron cuantas solicitudes había habido para el Programa Pareer y para el JESSICA. De las casi 400 solicitudes para el primero, no se habían aprobado ni la mitad.
Acabo el debate con nuevas felicitaciones y solicitud de agilidad en la puesta en marcha de las leyes expuestas.

Session 4:

How to obtain zero-energy buildings: maintenance?

Chairperson:

Vidal De Puig, Jordi

INCASÒL

Speakers:

Morer, Lluis
ICAEN. Catalonia.Spain.
Reynolds, Fabian
PROMUSA. Spain.
Valls, Gerard
Agència Catalana d l'habitatge. Catalonia.Spain.
Almécija, Joan Carles

ENERGEA, S.L.. Spain.

Conclusions:

Move to a performance based maintenance contract, where the energy performance of building is a parameter to maintain along the life spam of the building.
The key to achieve nearly zero energy buildings is to promote maintenance companies or Energy Services Companies in order to keep the energy savings over time
A new strategic energy model for energy contracts is necessary, in order to allow the renewable energies or high efficient energies to be integrated in buildings, with or without grid injection, and therefore, achieve energy and environmental targets.
Promoters of subsidized housing have the challenge of offering worthy homes with low energy needs, provided with low cost of maintenance systems, in buildings with the capacity of producing energy, and spending it in an efficient way
Energy saving and treatment are the clue to a sustainable world so it's absolutely necessary that experts (ESCO) drive them so that users can leave part of their concern: the part that is related to maintenance
Spain energy laws have to evolve; in Catalonia we have lack of opportunities right now.

Session 5:

Financial Structures and Incentives: How do we scale-up retrofit?

Chairperson:

Sweatman, Peter

Chief Executive, Climate Strategy & Partners, Spain

Speakers:

Title: On-going work in Europe led by the European Commission and its Energy Efficiency Institutional Finance Group to look at the barriers to attracting finance for energy efficient renovation

Dorendorf, Bettina

DG Energy, European Commission, Brussels, Belgium. Brussels.Belgium.

Title: Innovative financial instruments being developed for funding renovation and delivering deep renovation.

Bornholdt, Martin

CEO of DENEFF, the German Corporate Energy Efficiency Initiative. Germany.

Title: The policy and regulatory framework being evolved in Europe to help scale-up national renovation programmes

Lynham, Barry

Group Director of Strategy and Communication, Knauf Insulation. Belgium.

Title: The potential suite of other innovative financial solutions that can be put in place alongside private finance to ensure businesses and individuals take up renovation packages.

Lesay, Ivan

Advisor to the Vice-Chairman of Government and Minister of Finance. Czech Republic.

Conclusions:

Speakers presented a diagnosis of the current retrofit situation in the following terms:

1. Significant amounts of the funds needed, but the aggregate amount is not the current bottleneck.
2. Structures and incentives which drive demand for retrofits need to be prioritised.
3. Poor communication, lack of tailored finance and FI expertise are issues (in Germany);
4. Having funds available does not (by itself) create demand
5. All structured funds can be levered by the criterion of special investment vehicles or institutions to blend with private finance for energy efficiency.

Together with the audience, the solutions proposed to scale-up retrofit were:

1. Segment specific solutions are needed (residential, public, commercial, buildings) together with tailored FIs:
2. Building Renovation Financing Solutions need a “sales force” and to be distributed through retail bank networks (like KFW)
3. Renovations all need to be “energy efficient” and all opportunities to engage owners already renovating to move to deepen their renovations to save more energy should be

incentive.

4. Accounting regimes, especially for public banks and authorities, should not unfairly penalize energy performance contracts.
5. Strong policy leadership is required and present solutions alone will not deliver targets.

Could we consider...

6. An EU wide KFW-model that could deliver a trusted and standardized final product and distribution channel through all EU on leading retrofit banks and energy experts
7. New builds must be done as efficiently as possible.

Session 6:

Architecture based on circular economy, is this the right way?

Chairperson:

Llebot, Josep Enric

Current Secretary of Environment and Sustainability of the Government of Catalunya.

Speakers:

Braungart, Michael
Founder of EPEA International Umweltforschung GmbH in Hamburg, Germany, and co-founder of MBDC McDonough Braungart Design Chemistry in Charlottesville, Virginia, Dr. Braungart is currently a professor of Process Engineering at the University of Applied Sciences in Suderburg.. Germany.

Diaz Camacho, Miguel Angel

Architect, Ph.D. in Architectural Design, UPM ETSA Madrid. Madrid.Spain

Conclusions:

Las administraciones a través de la UE están implementando políticas hacia un modelo de economía circular. El camino es largo, pero la estrategia política está definida a implantar un modelo más sostenible. En cuanto a la implantación de un modelo similar en la arquitectura existen distintos grados.

En este caso hemos visto dos modelos de plantear la sostenibilidad en arquitectura. El primero se basa en cerrar el ciclo de materiales para eliminar o reducir impactos. El segundo excluye la posibilidad de hablar de residuos y argumenta a favor de "construir edificios como árboles", de generar impactos positivos en el que por ejemplo los edificios sean capaces incluso de mejorar la calidad ambiental interior.

Un único objetivo, pero varios caminos para conseguirlo.

Session 8:

Do we really want to build in the most sustainable manner? How the use of concrete may help for make sustainable buildings?

Chairperson:

Peñin Llobel, Alberto

Founder Peñin Arquitectos

Speakers:

Ramos Abengózar, José Antonio
Partner of Vicent & Ramos Estudios de Arquitectura. Spain.
López Antich, Marcos
Managing Director, Escofet 1886, S.A.
Arroyo Portero, Juan Carlos
Founder CALTER Ingeniería S.L.
Alonso Ollinger, Pedro

Housing Director, CEMEX.

Conclusions:

Las cuatro intervenciones de prescriptores (ingenieros y arquitectos), industrial y fabricante han puesto de manifiesto la necesidad de la investigación en una producción más sostenible del material base: el cemento. Algo que Cemex desarrolla en distintas líneas incluyendo la reposición del CO_2.

Siendo la construcción estimada en el 10% del gasto energético y el 90% la vida útil, la segunda pregunta del Congreso – la primera es obvia – subraya las cualidades del hormigón a lo largo de la vida útil: durabilidad (igual a sostenibilidad), eficiencia energética, reducción de huella ecológica, etc. La apuesta por la calidad, la racionalidad del proyecto y la capacidad integradora del material subrayan la respuesta afirmativa y abren caminos para el cumplimiento de los objetivos.

Session 9:

How to incorporate sustainable tropical timber in construction? Advantages, challenges and opportunities

Chairperson:

Rodríguez, Basilio

Fundación COPADE - Madrid - España

Speakers:

Nevado, Miguel
Enmadera. Madrid.Spain.
Alfaya, Valentín
Ferrovial. Madrid.Spain.
Anguita, Gonzalo
FSC Spain. Madrid.Spain.
Noriega, Ana Belén
PEFC Spain. Madrid.Spain.

Conclusions:

- We need to simplify the schemes of certification for the prescribers
- We need to create specifications for timber materials for prescribers
- We can build whatever with wood
- We must promote the use of woods in building.

Session 11:

How to harmonize building assessment at European level?

Chairperson:

Berchtold-Domig, Markus

Network Enterprise Alps (NENA), Schwarzenberg, Austria

Speakers:

Moro, Andrea
iiSBE Italia, Turin, Italy.
Borak, Dalibor
Czech Chamber of Architects. Prague.Czech Republic.
Viennot, Etienne
Rhône-Alpes Regional Energy and Environment Agency. Lyon.France.
Moosbrugger, Robert

Regional Development Vorarlberg. Alberschwende.Austria.

Concusions:

The main problem in Europe is that there are so many different ways, methodologies, tools and indicators to evaluate buildings. This problem causes difficulties to compare the efficiency between them (LEED, BREAM, etc..)

The effort that CESBA is doing is to try to legalize a parallel certification called "Building European Signature" so that some core indicators are common to different tools and therefore the results of them can be compared.

As a conclusion Governments should motivate people to certificate, with criteria and indicators that are easily understand not only by professionals but also for users.

Session 19:

Is invisible architecture the secret to happy, productive and healthy buildings? Specifically - acoustic comfort a privilege or a must

Chairperson:

Treasure, Julian

The Sound Agency, England

Speakers:

Daumal i Domènech, Francesc
Ph.D. in Architecture and Full Professor at ETS d'Arquitectura de Barcelona (UPC).
Czerwinska, Dominika

Manager of Membership and Europe Network, World Green Building Council.

Conclusions:

Hopefully we will have increased awareness of the need to design with sound.

Session 20:

Which are the keys to define roadmaps to sustainable building?

Chairperson:

Sweatman, Peter

CEO and Founder of Climate Strategy. Madrid, Spain

Speakers:

Garcia-Audi, Pau
Project Officer at European Commission. Brussels.Belgium.
Hidalgo, Andoni
Manager Comunications & Public Affairs Eurima. Brussels.Belgium.
De Santiago Eduardo
Technical Advisor. Deputy Directorate for Urban Planning. General Directorate for Architecture, Housing and Land. Ministry of Public Works. Madrid.Spain.
Joyce, Adrian

Secretary General EuroACE. Brussels.Belgium.

Conclusions:

Speakers provided the following diagnosis and introductory remarks:

1. Start from a shared definition of "sustainable building" with energy as a core vertical
2. Commission offers a mine step interactive JWG report detailing best practice from vision to delivery as well as various funding sources.
3. Energy Efficiency Directive is an opportunity to frame a transformational sector and the long-term roadmap (art 4) is key to deliver energy savings.
4. Roadmaps must contain
   - Long term view
   - Clue targets
   - Broad involvement
   - Holistic approach
5. Spain illustrates this process well, building on existing research, 21 stakeholder meetings and inter ministerial coordination to deliver new laws and finance
6. Detailed technical work driven form the physical reality and consensus solutions are key as shown in Spain experience
7. Buildings are core EU infrastructure, numbering 210 million, covering the land area of Belgium, and using 40%of primary energy
8. Energy savings of 131-192 billion euros by 2030 are possible if building renovation

roadmaps are successful, with the creation of 1-2 million direct jobs, and within 4-5 million indirect jobs.

9. Multiple considerable benefits of sustainable buildings include health, productivity, air quality, and asset value

Audience, together with the panel experts, concluded the following keys to resolve:

1. Long term buildings roadmaps need to be worth more than the paper they are written on, and the urgency of the delivery needs to be shared between policy makers, and the sector stakeholders, industry, property, owners, etc.
2. Individual buildings roadmaps can help owners to determine the best future evolution for their premises, and help them introduce the concept of a no-lose staged renovations.
3. Getting cross-Ministerial engagement and commitment are core to ensuring policy alignment
4. Explaining the benefits of renovation to the final user is critical and they need to also appreciate the multiple benefits of renovation. Communication strategies need to take care not to patronise nor fall into the "hoover" trap.
5. Accessible database of aggregate buildings technical passports need to be maintained to allow benchmarking and tracking of multiple benefits.
6. NZEB and future sustainable buildings strategies are critically connected to renewables and distributed energy models in Europe.
7. Everyone in the EU pays 2.75 € per day to import energy **– something needs to be done.**

Session 21:

Which methodologies, tools and projects for Energy Renovation in Buildings within MED Regional Strategies should be used in the framework of EU Directives?

Chairperson:

Marti, Xavier

Coordinator Marie Project. DTS-Generalitat de Catalunya. Barcelona. Spain

Speakers:

Cervelli, Curzio
Responsible for Project and Programme Development Unit. JTS MED PROGRAMME. Marseille.France.
Peters, Christoph
Architect. LIMA. Barcelona.Spain.
Ortiz, Joana
Researcher. IREC.
Castillo, Esther
Technical studies board. AVRA-Junta de Andalucia.
Herrenschmidt, Aniela
Chargée de project MARIE. Région PACA .
Llisterri, Fabian
Tresorery. UIA.
Abrigo, Luisa
Project officer. Piedmont Region.
Navarro, Pere
Cap Àrea Aprofitaments Fusters i Biomassa. CTFC.
Escobar, Juanjo
Head of division Energy Management. ICAEN-Generalitat de Catalunya.
Valentinčič, Miha
Head of estrategic development. PETROL D.D..
Papotis, Thanos
Tehcnical consultant of ANKO. EUROCONSULTANTS, S.A..
Boglio, Denis

Director general. CTFC.

Conclusions:

The following conclusions have been agreed by the audience.
Mr. Cervelli, Joint Technical Secretary of MARIE project, stated the great effort realised in the 3 years of operations, and also the overcome problems MARIE has gone through.
Mr. Martí, chairman and moderator, stated the current situation, enhancing the 3 MARIE drivers: policy, commitments portfolio, the MARIE product box, an idea embracing all the tools developed by MARIE partners and, finally, the Maximum Interest and Impact Project: the MIIP, which are the tangible produce of MARIE; projects that serve as tractor projects to start

and set the main MARIE target: a chain reaction in the market and the society to implement the Energy Renovation in Building policies.
The session is divided into 3 main blocks, each of them explaining the main works that MARIE partners have developed on each of the 3 MARIE Drivers:

1. Regional Strategies, reflecting the Policy Portfolio driver. Here, Mr. Martí explained the current 11 commitments already signed officially, but also the target of extending in the short term, this extent to the rest of the Ljubljana Declaration signers.
2. MEDBEES: Mediterranean Building Energy Efficiency Strategy. This point of the session explains the second point in the drivers cycle: the Product Box and its 4 main blocks: Intervention factors, Innovation Lab, Marketing and Training.
3. MIIPS: This block is focused on explaining some of the main MIIP projects that have been developed until this point of the project and which is their effect on the rest of the policies to implement replicated projects and improve results and achievements. Towards the 2020 objectives settled by the EU.

Session 23:

How can a demonstration project trigger the energy retrofitting of social and public housing in Europe?

Chairperson:

De Las Cuevas Jiménez, Juan Ramón

ACCIONA Infraestructuras S.A., Madrid, Spain

Speakers:

Alexandersson, Ulf
Alingsåshem AB. Alingsås.Sweden.
Jacobs, Anne
Luwoge Consult GmbH. Ludwigshafen.Germany.
Martinsson, Linda
Skanska Sverige AB. Göteborg.Sweden.
Hiller, Carolina

SP Technical Research Institute of Sweden. Borås.Sweden

Conclusions:

- We all agree that the consideration of social, economic and ecologic factors should be considered.
- The construction company, the contractor and the building users should have fluent and permanent communication, and a common objective.
- Barriers to covercome:
- Energy costs transfer
- Local and national regulation
- Privacy issues
- Tenant's allocatin during renovation
- The best way to trigger energy retrofitting is using real examples of real projects like BEEM UP, to prove that it can be well worth it.

Session 24:

What is the right approach to energy and natural resources in the sustainable cities?

Chairperson:

Manteca, Florencio

MSc Architect. Director of the Energy in Buildings Department at the National Renewable Energy Centre of Spain – CENER.

Speakers:

Treberspurg, Martin
Univ. Prof. Arch. DI Dr, Head of Division of Sustainable Construction, University of Natural Resources and Life Sciences,. Vienna.Austria.
Astibia, Alejandro
Dipl. Ing. Manager at Energy Agency of Pamplona, Spain. Pamplona.Spain.
Roux, Saul

ACC - African Centre For Cities - Cape Town, South Africa. Cape Town.South Africa.

Conclusions:

- Need to define an integrated and holistic approach, the different aspects to define the urban activity: urban planning, urban regeneration, mobility, demand side management and also energy supply.
- The scenario of a self-sufficient city is a goal that we should have in mind and may be a reality at medium term
- Need to re-define the relation between big cities with the smaller villages nearby.

Session 26:

How can innovative technologies make a difference in building retrofitting? The RESSEEPE approach

Chairperson:

Centeno, Fernando

Head of research projects (Exergy)

Speakers:

Roman, Eduardo
Solar Photovoltaic Manager in the SOLAR Area of Energy and Environment Division (TECNALIA). New technologies for refurbishment in RESSEEPE.
Sakas, Nikkos
Manager and in change of the WIRELESS THINGS spin off activity under buildingwarden.com (Apintech Ltd). Innovative sensor development and monitoring in RESSEEPE.
Pérez, Patricia
Architect in Technical Department of Building Projects (OHL). New technologies applied to building demosites in RESSEEPE.
Ferrando, Valeria

EU Head of Research (IES). Numerical simulation of buildings in RESSEEPE.

Conclusions:

- In general, we have presented the main objectives of the RESEEPE project, where we have highlighted the benefit of integrating a balanced mix of innovative and state of the art technologies in order to achieve energy savings of up to 50% in the retrofitting of public buildings. We have gone through several aspects: project objectives, technologies, pilot implementation, result validation and simulation approaches.
- We have had some discussion on innovative technologies such as innovative thermal solar collectors with improved efficiency. Discussion on simulation has been very productive, where the speaker has highlighted the challenge of modelling new technologies (electronic-photovoltaic windows and PCM), for the simulation of energy efficiency of public buildings.

Session 29:

How can ICT technologies be relevant enablers for Energy Efficient Buildings?

Chairperson:

Griera-i-Fisa, Mercè

European Commission, Communications Networks, Content and Technology Directorate-General, Smart Cities & Sustainability. Brussels, Belgium

Speakers:

Jiménez Redondo, N.
CEMOSA, Málaga, Spain.
Amorós, F.
Universitat Politècnica de Catalunya, Department of Construction Engineering. Terrassa.Spain.
Lanceta Alonso, D.
Solintel M&P, SL. Madrid.Spain.
Costa, A.

Informatics Research Unit for Sustainable Engineering, National University of Ireland,. Galway.Ireland.

Conclusions:

Definitively ICT is an enabler for energy efficient Buildings.
Issues that require further attention are:

- Interoperability to avoid vendor lock-in
- Value for money. Fast recovery of investment
- Comfort and citizens acceptance. There is a EU standard on comfort
- A transparent measurement methodology to ensure comparability of the saving achieved. IMPV could become a standard
- Reliability. The system should be able to survive even if a few sensors fail
- Industrialisation of the construction sector.

Session 37:

Is it possible to achieve an alternative real-estate model where "nearly" everyone wins?

Chairperson:

Bermejo Amarillo, Javier

Marketing and Kömmerling Prescription Manager

Speakers:

Mira Illana, Miguel Ángel
Arquitecto, Socio fundador y Presidente de la Asociación Jubilares. Socio Fundador de Milla Mira y Navarro Arquitectos S.L.
Style, Oliver
Cofundador de Progetic y Director del Área de Edificios de bajo consumo. Consultor energético, Proyectista Passivhaus Certificado.
Robert, Raul
Industrial Engineer, Socio Fundador Sostre Civic.
Ruíz, Rogelio

Arquitecto, eCOHOUSING Equipo Bloque Arquitectos.

Conclusions:

La respuesta es “sí”, efectivamente existen modelos alternativos, viable económicamente y sostenibles.
Aquellos en los que el usuario está en el centro de la toma de decisiones y participa en él desde el minuto cero.
Aquellos en que se contempla el proyecto en el concepto del Ciclo de Vida del edificio, especialmente en su etapa de uso, teniendo en cuenta la eficiencia energética.
Ejemplos como los modelos de cooperativas de cesión de uso, *co-housing*.
Solamente pierden los que tienen establecido su modelo en la “ESPECULACIÓN”.

Session 38:

Are sustainable buildings worth the effort? (The view from various sectors involved)

Chairperson:

Lechien, Dominique

Partner – Chairman of Baker & McKenzie Global Real Estate Practice Group

Speakers:

Alfaya, Valentín
Quality & Environment Director - Ferrovial.
Reihlen, Albrecht
Managing Director - Deka Immobilien Investment GmbH.
Amores, Miguel Ángel
Coordinator of the Environmental Department.
Hirigoyen, Julie

UK Head of Sustainability -Jones Lang Lasalle.

Conclusions:

Yes, sustainable building is worth the effort for most of the buildings for various reasons.

- Improve value
- Reduce impact on environment
- Reduce cost and energy dependency
- Improve efficiency
- Improve social wellbeing of workers
- Improve corporate image.
- Adapt legislative trends and work out production evolution

Session 39:

How can we put in practice a multilevel Governance model integrating private sector for energy renovation in buildings in Mediterranean?

Chairperson:

Strangis, Davide

Executive Secretary, CPMR Intermediterranean Comission

Speakers:

Maldonado, Eduardo
Coordinator of "European Concerted Building Action". University of Porto. Porto.Portugal.
Cervelli, Curzio
Responsible for Project and Programme Development Unido -JTS Med Programme.
Marti, Xavier
Coordinator MARIE project. DTS-Generalitat de Catalunya. Barcelona.Spain.
Crotta, Stefania
Deputy Directore Regionalle 21. Regione Piemonte. Regione Piemonte.Italy.
Tourret, Jean Claude
Directeur. AVITEM, ELIH-Med partner, France.
Naguib, Amin
CES MED Coordinator. Tunisia.
Laverdière, Folco

Chef de service adjoint. Région PACA.

Conclusions:

- Boosting the coordination among key players (in the broad sense) at public and private level , through clusterization, joint programming, capacity building, exploiting the opportunities of the new financial framework and in particular programmes like MED/ ENPI /H2020 and ESSF FUNDS and the ones of future regional strategies.

- The base should be the one of regional strategies on ERB.
- JAP- / 5 axis/- 5 axis proposed – brainstorming further for the implementation.
- Policy paper – 28/11 Final event ELIH-MED. Brussels participation of EP.
- New way: proximity:
    - north has to understand
    - Message≠taking into account
    - ≠ political structures

Session 40:

5th Active House Symposium, Session 1: How can sustainable buildings become standard solutions and affordable for people and society?

Chairperson:

Eriksen, Kurt Emil

Secretary General, Active House

Speakers:

Title: The European Commission's view on sustainable buildings and future steps to strengthen the focus on sustainable buildings.
Garcia, Pau
EASME - DG Energy European Commission.
Title: Can we use learnings from the past to develop standard solutions for sustainable buildings?
Larsson, Niels
Executive Director. International Initiative for a Sustainable Built Environment (iiSBE).
Title: What are the value of sustainable buildings and how can we engage customers to invest in sustainable buildings?
Haglind, Staffan
International Green Business Officer. Skanska.
Title: How can the industry support a development towards sustainability and sustainable living?

Rasmussen, Michael K.

Chief Marketing Officer VELUX A/S and Chairman of Active House Advisory Board.

Conclusions:

1. Generally speaking, we need to strengthen the awareness of all the values of sustainable buildings including the value of better indoor climate and the higher productivity and health for people living in sustainable buildings. And we need to make those values visible for investors and occupants.
2. We should also use the methodologies available already today, and architects, investors and occupants must set targets for sustainability already in the early design process, in order to use those targets as dialog tools during the design process, as well as it shall avoid higher costs later in the design process.
3. Finally we shall focus on both new buildings and climate renovation of existing buildings, and we shall see buildings as an element in the city and neighbourhoods, and we should avoid sub optimization on specific topics, like integration of renewable energy in buildings, which can be more sustainable if it is installed in the grid. Sustainability must be evaluated on city level or even larger scale.

Session 41:

The future of Solar Decathlon Europe?

Chairperson:

Rollet, Pascal

Competition Director of SDE 2014, Versailles

Speakers:

Jacques Le Seigneur, Vincent
INES. France.
Mesureur, Bruno
CSTB Solar. France.
Rodríguez, Edwin
CSTB Solar and Universidad Politécnica de Madrid. Madrid.Spain.
Veaga, Sergio

CSTB Solar and Universidad Politécnica de Madrid. Madrid.Spain.

Conclusions:

SDE2016 is called by all participants.
There is a need for new governance at EU level in order to guarantee stability.
The event should be opened to evolutions regarding ranking and evaluation.

Session 42:

Can Sustainable Resources Management Keep Up With Rapid Urbanisation / Building Construction?

Chairperson:

Poon, C.S.

Professor, Associate Head and Director of Research Centre for Environmental Technology and Management at the Civil and Environmental Engineering Department, The Hong Kong Polytechnic University; Hong Kong Special Administrative Region (HKSAR)

Speakers:

Title: Innovative Design Measures for Waste Minimisation in Public Housing Development
Fung, Ada Y.S.
Deputy Director, Hong Kong Housing Authority, HKSAR.
Title: Moving Towards Low Carbon Building Design and Construction – From Rhetoric to Reality
Lee, Chun Fai
Manager - Research, Construction Industry Council, HKSAR.
Title: Case Studies of Development and Application of Green Concrete in Hong Kong
Zheng, W and So, K.L.
Gammon Construction Ltd., HKSAR.
Title: Commercialisation and Applications of Recycled Construction Materials

Chan, Dixon

Tiostone Holdings Ltd, HKSAR.

Conclusions:

Conclusion: yes, but subject to the following conditions:

1. There must be concerted efforts between governments and the private sector in promoting the use of green materials
2. Suitable guidelines and specifications be established to ensure quality
3. Pioneers in the use or development of green materials should be recognized
4. Life cycle cost should be considered rather than only selling price.

Session 43:

What is the role of construction materials suppliers to contribute to more sustainable buildings?

Chairperson:

Hofmann, Dr. Matthias

Sika AG, Zurich, Switzerland

Speakers:

Alonso, Carmen
IETCC-CSIC. Madrid.Spain.
Zambrana-Vasquez, David
CIRCE. Zaragoza.Spain.
Carmona, Ana
Sika, S.A.U.. Madrid.Spain.
García de Oteiza, Rafael

Grupotec. Barcelona.Spain.

Conclusions:

- Research in construction materials + application is key
- Comparative analysis of building materials is necessary
- Embodied energy and use phase energy consumption should be balanced
- There are materials already existing which are working for more sustainable
- Long lasting building materials
- Education
- Commitment + financial incentives

Session 44:

How to approach the project, with a common global commitment, from different environmental, social and economic al situations? The transformation in the architecture from the complicity between students and teachers

Chairperson:

González, María Jesús

ASA President (Association of Sustainable Architecture) from 2010 to 2013. Member of the Scientific Committee of the contest POWERING TRANSFORMATION

Speakers:

Title: CO3-Cooperativeness for Sustainable Regeneration. *First Award*.
Moreno, Susana
European University of Madrid. Madrid.Spain.
Title: CO3-Cooperativeness for Sustainable Regeneration. *First Award*.
González, Francisco Javier
European University of Madrid. Madrid.Spain.
Title: WATERNET-La Yaguara resilience park. *Second Award*.
González, Yumisbet
Simon Bolivar University. Venezuela.
Title: WATERNET-La Yaguara resilience park. *Second Award*.
Piscitelli, Isabel
Simon Bolivar University. Venezuela.
Title: PLUGGET MOBILITY-Co-working facilities in Barcelona. *Third Award*.

Izaga, Pablo

Sevilla University. Sevilla.Spain.

Conclusions:

- The approach must be global: the importance of an integral and correct diagnosis is crucial
- It is necessary to have and to develop new methodologies to obtain the global point of view. Other basic necessities:
- Involving citizens
- Improve the schools of Architecture connexion with real problems
- The 5 criteria of the Powering Transformation competition are really good and must be spread (1, cross-cutting scale; 2, horizontality; 3, multidisciplinarity; 4, interactivity; 5, innovation).

Session 45:

How does the sustainable use and operation of public buildings in Germany works?

Chairperson:

Lützkendorf, Thomas

Head of the Chair of Sustainable Management of Housing and Real Estate Department of Economic and business Engineering at Karlsruhe Institute of Technology (KIT)

Speakers:

Müller, Julia
Federal Institute for Research on Building, Urban Affairs and Spatial Development (BBSR), Berlin, Germany.
Leide, Lutz
Institute for Federal Real Estate (BImA),. Bonn.Germany.
Schablitzki, Gerd
Federal Environment Agency (UBA). Dessau.Germany.
Wagner, Andreas

Karlsruhe Institute of Technology (KIT), Karlsruhe, Germany.

Conclusions:

- The German federal Government is deeply committed to sustainability in the building environment. Due to the dominance of older buildings, the BNB assessment system has a new module for existing buildings. The largest federal building management organization, BImA, currently uses an environmental management structure called LUMAS, a holistic approach. The German Environmental Ministry, an agency of the Federal Government, uses EMAS as a tool for providing overview and direction in both reducing resource consumption and increasing user comfort. There is an increasing priority on creating a feedback loop with building users and occupants to provide a better workspace in buildings.

Session 46:

How can planners and investors benefit from best practice projects on sustainable buildings?

Chairperson:

Vasallo, Ali

Fundación CARTIF, Boecillo, Valladolid. Spain

Speakers:

Zabalza, Ignacio
Fundación CIRCE. Zaragoza.Spain.
Kaiser, Jan
Fraunhofer-Institut für Bauphysik IBP. Kassel.Germany.
Lalaguna Alcaine, Iván
INGECON (Ingeniería y Control Electrónico). Zaragoza.Spain.
Sanz, Sergio

Fundación CARTIF. Boecillo, Valladolid.Spain.

Conclusions:

- Planners and investors should be involved from the very beginning in the process of developing sustainable buildings considering an IPD approach and tools like BIM, LCA and LCC.
- Planners and investors should consider an integrated holistic approach in order to maximise comfort and energy efficiency
- Understanding how end users use the building is crucial for a proper building design
- Sustainable buildings require the new business models involving all stakeholders
- Sustainable buildings are 100% marketable
- There is no universal solution

Session 57:

Mediterranean region. What challenges will the methodology of Energetic Renovation of LogementS (RELS) have to tackle with?

Chairperson:

Claret, Coque

Architect and Technical coordinator of RELS project at University UPC, Barcelona, Spain.

Speakers:

Title: Opening and Welcome of Housing Agency of Catalonia
Rodríguez Marín Sastre, José Pablo
General Manager, Housing Agency of Catalonia. Spain. Spain.
Title: Opening and Welcome of Politecnic University of Catalonia
Puigdomenech, Joan
Vice-rector, Comunity, Social Action and Sustainability UPC. Spain.
El Ghosseini, Nouha
Mayor of Baakline. Baakline.Libano.
El Hanchi, Fethi
Director, Rational use of energy division, National Agency for Energy Conservation (ANME). Tunisia.
Watson, Ian

Expert, BRE Group, Liverpool Warm Zone. United Kingdom.

Conclusions:

- La conclusion principal:
- Se deben abordar los problemas de eficiencia de forma multiescalar y en mesas multidisciplinares
- Seguramente afrontar las políticas de forma más práctica: de la persona, al edificio, al barrio y a la ciudad
- El papel de la Universidad es clave, pero hay mucha investigación aprovechable
- Los cambios de relación respecto a las políticas obligan a mesas más ricas de diálogo.

Session 58:

5th Active House Symposium, Session 2: Lesson and experience with design and evaluation of Active House

Chairperson:

Gunther, Zsolt

Architect, 3H Architect. Board of directors Active House Alliance

Speakers:

Title: How do sustainable buildings affect the behavior and well-being of occupants?
Wegener, Bernd
Humboldt University Berlin. Berlin.Germany.
Title: Initiatives to move towards sustainable homes in UK
Halsall, Peter
Sustainable Developer, CEO of Good Homes Alliance. London.United Kingdom.
Title: Use of the Active House Vision in development of specification to Green Solution House, a renovation and extension of a conference center
Richter, Trine
CEO Green Solution House. Ronne.Denmark.
Title: Learnings from practical application of the Active House Specifications and Active House Radar: Suggestions for possible improvements
Holzer, Peter
Institute of Building Research & Innovation. Vienna.Austria.
Title: Learnings from practical application of the Active House Specifications and Active House Radar: Suggestions for possible improvements

Hammer, Renate

Arch. Dipl-Ing.Dr; Institute of Building Research & Innovation. Austria.

Conclusions:

- An easy-to-explain graphical communication (E.g. with the radar diagram) in the language of the end users with an exact calculation behind it could help in increasing the number of sustainable buildings.

Session 59:

Can low-cost financing achieve bankability of large-scale deep energy retrofit in public housing?

Chairperson:

De Astorza, Carlos

AVS, Spanish federation of public housing companies

Speakers:

Bastiaanssen, Rolf
Bax & Willems Consulting Venturing.
Dorendorf, Bettina
DG Energy, European Commission/Energy Efficiency Finance Institutions Group.
Cole, Steve
National Housing Federation (NHF), United Kingdom.
Russell-Croucher, Martin

RICS: Royal Institution of Chartered Surveyors, London, United Kingdom.

Conclusions:

The main ideas exposed in the session as proposals to overcome the challenge of Energy Efficiency financing in the social and public residential sector in EU are:

- EU Housing Fund - creation and risk sharing.
- New schemes considering long-term investments and strategies adapted to it.
- Standardization of processes to design a common EU methodology, considering the specificities of each member state.
- Combination of financing instruments to design a product that guarantees the implementation of Buildings European Energy (BEE) strategy.

Session 60:

Is Urban Compaction our Sustainable Future? - The Hong Kong´s Experience

Chairperson:

Poon, C.F. Larry

Architect planner in private practice with the Hong Kong-based consulting firm, Wong Tung and Partners Limited. He is currently a Committee Member of the Hong Kong Green Building Council and a Convener steering a consultancy study in developing a new rating tool for planning applications for the Council; Hong Kong Special Administrative Region (HKSAR)

Speakers:

Title: Urban Regeneration Focusing on Community Cohesion
Tam, S. Y. Iris
Managing Director, Urban Renewal Authority, HKSAR.
Title: Culture & Place Making: West Kowloon Cultural District Development
Chan, M. W.
Executive Director of Project Delivery, West Kowloon Cultural District Authority, HKSAR.
Title: Accessibility & Connectivity: Mass Transit Development
Cheng, K. W. Ronald
GM - Technical & Engineering Services, MTR Corporation Limited.
Title: China's New Urbanisation Plan 2014-2020: Containing Urban Sprawl and Mitigating Carbon Emissions

Yip, Stanley

Director of Planning and Development, Arup China, Shanghai, China. School of Urban Planning and Design, Peking University, Beijing, China.

Conclusions:

About 200 participants attended the session, 11 of them voted "yes", 10 voted "no" and 18 voted "yes but with qualifications" to the session question. That is, a majority of the audience is for the question but certain reservations such as the following:

1. Land supply is city core: Intensification of land uses can only take place if there is a continuing supply of land or old building blocks for city redevelopment
2. Urban governance / investment: the ability of a city to improve liveability relies largely on its governance, the availability of a mature sophisticated planning and urban management system at work
3. Respect and openness: to make a compact city work, people must respect the freedom of others in very limited space and observe social order.

Session 61:

Can the building sector contribute to mitigate climate change?

Chairperson:

Pedrals, Nuria

President of AUS

Speakers:

Samitier, Salvador
Climate Change Office Head. Spain.
Caltayud, Dani
ETSAV professor. Barcelona.Spain.
López, Fabián

Architect PhD, Societat Orgànica. Barcelona.Spain.

Conclusions:

Salvador Samitier:

- Última década 2,2% anual doble incremento de emisiones
    - Nivel mundial incremento PIB
    - Menos intensidad energética
    - Menos descarbonización energía
    - Población ≈
- 2030: reducción 30% en emisiones respecto al 2005
    - 27% renovables
    - 27% eficiencia energética (Cataluña 20%)
- Objetivos y criterios mínimos, cuantificación emisiones (ley de cambio climático) (no suficiente)
- Energía verde Infraestructuras publicas, autogeneración, …
- Información publica. Fines de año.

Fabián López:

- Hay herramientas para calcular las emisiones de los edificios. Construccion y consumos.
- Consumo referencia 8.000 kw/m2 año
- Nueva union del sector, circuito cerrado…

¿Como se define el consume casi nulo?
Forma de medir la compacidad o densidad para habitar la ciudad
¿Exterminacion de la ciudad?

Session 62:

Powering transformation. Which is the value of the intangible in the architecture and the sustainability?

Chairperson:

Díaz Camacho, Miguel Ángel

ASA President (Association of Sustainable Architecture)

Speakers:

Pich-Aguilera, Felipe
Architect. President of GBC Spanish.. Spain.
Rich, Peter
Architect, founding partner of Light Earth Designs LLP, Honorary Fellow of the American Institute of Architects..
Lau, Stephen
Architect, professor of the proyects architecture school of the University of Hong Kong. Chair of the China-Hong Kong Green Building Council Hong Kong Chapter.. Hong Kong.China.
Elizalde, Antonio

Sociologist, member of Sustainable Programme of Chile.. Chile.

Conclusions:

1. Intangibles are really difficult to measure: how can we measure the sense of community, unskilled labour, environmental education related to architecture and design, Do It Together processes or democracy decisions?
2. There are some cultural issues (lifestyle) before architecture related to intangible. We need three basic environmental qualities for our life: health, safety and wellbeing. Do we need to quantify beauty? Is it possible?
3. Indicators don't qualify architecture
4. We can't measure everything. Are we measuring entropy? Ethics? Ecological policies? Are we measuring the impact of wrong regulations?
5. It's not possible to separate tangible and intangible, it would be a huge mistake to forget one of them. We need to develop some new methodologies for architectural design including tangible and intangible issues. We need to change the past general idea about what "good architecture" really is. Finally architecture is about ethics and true respect for human being.

Session 71:

Designing for hope: What does contributive practice for a thriving city look like?

Chairperson:

Hes, Dominique

The University of Melbourne

Speakers:

Du Plessis, Chrisna
The University of Pretoria.

Conclusions:

1. A clear shift is necessary in the way we conceive the built environment, from separated social and ecological systems to an understanding of the built environment and all other complex systems as integrated "living systems".
2. As an emerging concept within environmental design, regenerative design proposes a change in mindset to enable architecture to be a pin-pointed catalyst of change for the larger systems in which is nested, also enabling it to influence other important aspects of human life beyond building design and support all other living systems and natural resources cycles.
3. For this, some scholars see the role of building design needing to shift from the creation of eco-efficient buildings to the design of projects that encourage and support resources exchange, an inherent characteristic of all living systems.

Session 72:

What are the main barriers and opportunities for EcoCities in Egypt and similar countries?

Chairperson:

Hedman, Åsa

Senior Scientist, MSc (Tech). Project Manager of "EcoCity Capacity Building in NBC". VTT Technical Research Centre of Finland

Speakers:

Title: Industry sectors' role in the EcoCity approach in Egypt. Impacts, barriers and opportunities
ElShazly, Ahmed
Chairman, Chemical and Petrochemical Engineering Department Egypt-Japan University of Science and Technology.
Title: Energy sectors' role in the EcoCity approach in Egypt. Impacts, barriers and opportunities
Adel-Rahman, Ali K.
Chairman, Energy Resources Engineering Department Egypt-Japan University of Science and Technology.
Title: The view of city authorities and investors on the challenges and opportunities of EcoCities in Egypt
El Menshawy, Hany
Chairman of Institute of Industry Committee (IIC).
Title: The view of national authorities on the challenges and opportunities of EcoCities in Egypt

Moustafa, Eng Hoda

Representative of EEAA, Egypt Environment Affairs Agency, Ministry of the State of Environment.

Conclusions:

- Energy deficiency
- Financial barriers
- Business and informational barriers
- Permits, licenses and legal barriers
- Regulatory and legislation barriers
- Cultural barriers
- Lack of advanced technology

Session 73:

Sustainable architecture or common sense?

Chairperson:

Mangado, Francisco

Architect. Teacher at the University of Navarra. Pamplona, Spain

Speakers:

Carrilho da Graça, João Luís
Architect. Lecturer at the University of Lisbon.. Lisbon.Portugal.
Vazquez Consuegra, Guillermo
Architect. Honorary teacher at the University of Sevilla.. Sevilla.Spain.
Herzog, Thomas

Architect Professor of Energy Efficient and Sustainable Planning and Design at the Technische Universität. Munich.Germany.

Conclusions:

- Se está desarrollando con intensidad y extensión una preocupación por una arquitectura sostenible. Algo interesante. También una cierta banalización del término. En todo caso parece más lógico hablar de una arquitectura responsable donde el sentido común es importante y donde una adecuada relación entre medios y fines defina un contexto amplio en el que insertar toda la complejidad implícita en el ejercicio de la arquitectura.
- La sostenibilidad no puede significar solo una simple adición de elementos y componentes técnicos. Tiene que ser con algo más rico donde la investigación y el desarrollo de las ciencias de la construcción son instrumentos fundamentales. Se hace necesaria más arquitectura. Más reconocimiento y manejo de todas las realidades e instrumentos arquitectónicos utilizados con sentido de responsabilidad. Un mayor reconocimiento de la condición contextual, de la durabilidad, del sentido de servicio, de los justo y lo adecuado se hacen fundamentales. No solo las normativas minuciosas, en ocasiones contradictorias, son la solución.
- La solución tiene que ser con la investigación y mayor sentido de lo adecuado.

Session 74:

How to promote Nearly Zero Energy Buildings for new and retrofitting buildings in the municipal context

Chairperson:

Cipriano, Jordi

Director of CIMNE-BEEGROUP, partner of the AIDA project, Barcelona, Spain

Speakers:

Title: Powerhouse Kjørbo: The experience of a Plus-Energy office building in refurbishment
Sartori, Igor
Senior research of NTNU - Norwegian University of Science and Technology. Trondheim.Norway.
Title: Zero-Energy Urban Buildings and Zero-Energy Communities
Santamouris, Mat
Professor of NKUA - National and Kapodistrian University of Athens. Athens.Greece.
Title: Simulating intervention scenarios for carbon reduction in urban planning (SEMANCO Project)
Cipriano, Xavi
Researcher of CIMNE- BEEGROUP, Partner of the SEMANCO project. Barcelona.Spain.
Title: The nZEB promotion within SEAP (AIDA Project)

Santos Lopez Lorenzo, José

Researcher of CIMNE- BEEGROUP, partner of the AIDA project. Barcelona.Spain.

Conclusions:

Different points of view were boarded:

Strenghts

- Within high ambition retrofit strategies (e.g. passive house standard, innovative ventilation strategies) by using high-efficient materials and the optimization of energy supply systems for production of thermal energy and electricity onsite, there is a high possibility to achieve a zero-energy level.
- Zero-energy buildings and zero-carbon settlements are a huge challenge for the scientific community that already works towards advanced conservation techniques and passive solar technologies in order to reduce the issue of high energy consumption in the present building sector. If an appropriate strategy is consolidated, the building sector will be permitted to achieve a future less consuming, near zero-enery and more comfortable buildings.
- The simulation of intervention scenarios boosts the future development for urban plans with the nZEB achievement even more realistic and consistent than individually performed at building level.
- SEAP encourages public authorities to fulfill the commitment of 20% reduction in CO2 and promote the “nearly” Zero-Energy Buildings construction in the near future according to the energy performance of building directives.

Weaknesses

- Despite past and present research and pilot actions have been performed and can solve major energy and environmental problems, to assimilate them in the practice is always a slow process and it involves favourable mechanisms to achieve relevant results in the future.
- To pass from research to practice is always a slow process and requires a favourable social, financial, legislative and technical environment and to start with this process, a huge financial investment potential for retrofitting should be achieved.
- Simplified software tools are not accurate to perform reliably energy simulations and sometimes they can easily lead to error. Then, expert knowledge is required to modify any predefined data. For individual buildings, energy audits give more reliable results for the achievement of nZEB retrofitting.
- Spain is one of the countries there are more municipalities with SEAPs subscribed in the Covenant of Mayors. This reality seems nearer to be eligible for possible funding opportunities instead of reaching the target goals for reducing CO2 emissions or having in consideration the nZEB as worthwhile investment.

New developments towards nZEBs

- Project tools: Net ZEB Evaluation Tool for the energy balance, SEMANCO Platform for simulating future urban scenarios AIDA guidelines for tenders and roadmaps in the line of promoting nZEBs in the municipal or regional context.
- Simplified tools and methods: Official Methodologies to normalise Energy Performance Certifications, the new CEN standard 'FprEN15603 Energy Performance of Buildings' to define nZEB boundaries and Energy Management tools like SIE to collect all the possible building existing data in energy consumption, building features, etc.

Session 75:

Are current building regulations adequately advancing sustainable buildings? If not, what is missing, and how should they be changed? (I)

Chairperson:

Serra, Javier

Ministry of Infrastructure, Spain

Speakers:

Title: Resilience of Australian Buildings to Extreme Weather Events
McDonald, Mathew
Australian Building Codes Board. Australia.
Title: Life-time Environmental Sustainability of Buildings under the Singapore Building Control Act
Neng Kwei Sung, Jeffery
Centre for Sustainable Building and Construction, Singapore. Singapore .
Title: Sustainability Labeling for Building Standards
Dodds, Bill
Scottish Government. Scotland.
Title: Building Regulatory Control in Facing the Challenge of Climate Change: A Case of Hong Kong
Chan, Edwin HW
Hong Kong Polytechnic University. Hong Kong.China.
Title: Sustainability, resilience and risk in earthquake-prone areas: Lessons for building regulators from the Cantebury earthquakes

Stannard, Mike

Chief engineer, Ministry of business, Innovation & Employment (MBIE). New Zealand.

Conclusions:

La sostenibilidad en los Códigos debe ir más allá de lo ambiental, debe contemplar los aspectos sociales (accesibilidad, etc.) y económicos.
La sostenibilidad debe incluirse en todas las exigencias actualmente contempladas.
La normativa debería ser el último recurso para intervenir, o llevará aún más tiempo el que la industria se autoregule. Hay otras estrategias para alcanzar los mismos objetivos de manera efectiva.

Session 76:

What are the best "sustainable" construction materials?

Chairperson:

Eveillard, Pascal

Sustainable habitat and public affairs director, Saint-Gobain

Speakers:

Mesureur, Bruno
Board Member SB Alliance & director marketing and international development CSTB.
Lindblom, Josefina
DG Environment, European Commission.
Domínguez Martínez, Alejandro
R&D Manager, Saint-Gobain Placo. Spain.
Antink, Roy

Director Green Support, Skanska AB.

Conclusions:

- Sustainability for construction products cannot be looked at independently from the building performance. Therefore defining sustainable performance at building level first is key.
- The best products cannot deliver at building level if they are not properly installed. Quality of installation is necessary to deliver sustainable buildings
- Transparency as the product performance is more and more needed and the weight of products and materials in assessed schemes will continue to increase.
- These is a need for a common language and especially for common indicators, standardised, easy to use
- There is a challenge to make sure that the whole building chain is being empowered and that all stakeholders even if small will follow the trail.

Session 77:

Which breakthrough innovations will contribute most to acceleration of the sustainability transition in buildings?

Chairperson:

Bordeau, Luc

ECTP/E2BA Secretary General (European Construction Technology Platform/Energy Efficient Buildings Association)

Speakers:

Van Heiningen, Rinske
Akzonobel. The Netherlands.
McLean, Don
IES. United Kingdom.
Marti, Alain
EDF (France).
Bastiaanssen, Rolf

Bax & Willems Consulting (Netherlands). The Netherlands.

Conclusions:

Around 40 persons in the room. Very interesting various solutions for innovating and deploying them into the building sector have been presented by the speakers, with a focus on materials, ICT, connecting to the grids, and innovations process itself. Exchanges with the audience took place on:

- The fact that practitioners from industry were invited to give their opinions and views from the "real field".
- Living Labs and their certification.
- Replicating solutions is not always possible
- Skills of workers is also an important issue if not the first one to be sure than innovation is correctly implemented.
- Design activities are of major importance too. Errors (which can not be corrected later on) at this step have to be avoided, which is more and more difficult taking into account the increased complexity of the requirements and issues to be tackled.
- Importance of the value chain is crucial. Business models have also to be innovated. Performance guarantee systems are an example of such innovative business models.

Luc Bordeau

Session 78:

Zero Carbon Buildings and Beyond! What is the future for Sustainable Built Environment?

Chairperson:

Yau, M.H. Raymond

Arup Fellow and Director of Arup Hong Kong. He is Arup's director responsible for the total engineering design of Hong Kong first Zero Carbon Building of Construction Industry Council. He is Director of the Hong Kong Green Building Council and 1st Vice Chairperson of BEAM Society Limited

Speakers:

Title: Towards Next-Positive Energy Buildings
Cole, J. Raymond
Centre for Interactive Research on Sustainability, UBC. Canada.
Title: Implementation of Zero Carbon Buildings in Asia
Cheng, Vincent
Director of Building Sustainability, Arup Hong Kong. HKSAR.China.
Title: Towards Zero Carbon Building-Bridging the performance Gap
Chan, Cary W.H.
General Manager, Sustainability and Technical Services, Swire Properties, HKSAR.
Title: Smart Energy Systems for Smart Towns

Wang, Shengwei

Chair Professor of Building Services Engineering, The Hong Kong Polytechnic University, HKSAR.

Conclusions:

We have come to a conclusion with simple majority in support of the debate question: Zero Carbon Buildings or Energy Positive Buildings have a clear role to play in the future for Sustainable Built Environment. However, it will need to be considered in depth the following aspects:

1. Market drives or policies to deliver ZCB or beyond
2. Behaviour change to drive uptake of technology advancement to deliver performance
3. Institutional barriers to overcome
4. Extension to neighbourhood or district level by advanced infrastructure
5. Pricing strategy of feed-in tariff and cross-subsidies across electricity provides fed in to grid and those normal users.

Session 82:

Eco-Innovations in Value Driven Sustainable Building Processes Need for New business Models, Opportunities for SMEs

Chairperson:

Huovila, Pekka

VTT Technical Research Centre of Finland

Speakers:

Antuña Rozado, Carmen
VTT Technical Research Centre of Finland. Finland.
Lommi, Jukka
Tiuri & Lommi. Finland.
Trinius, Wolfram
Trinius GmbH. Germany.
Pinon, Ludovic

Lascom. France.

Conclusions:

The session had a focus on BIM which was not highlighted in the session description. Most of the audience weren't familiar with BIM and therefore no detailed conclusions were drawn together with the participants.
In general, opportunities for sustainable business models for SMEs were identified and as the panellists witnessed, some new tools already implemented.
A good session!

Session 91:

What's the potential of building well-designed, high-performance sustainable homes at a price low-income families can afford? By Make It Right non-profit organization.

Chairperson:

Tijeras, Valentín

Product and innovation Manager at Consentino Group, Cantoria, Almeria, Spain

Speakers:

Darden, Tom
Make it Right Foundation. New Orleans.USA.
Turner, Craig
Make it Right Foundation. New Orleans.New Orleans .
Williams, John

John C. Williams Architects, New Orleans, United States of America.

Conclusions:

Key factors:

- Use every challenge as an opportunity.
- Excel in everything you make
- Importance of community approval
- No charity, help people to rebuild their lives
- Use local talent
- Continuous innovation, every house is a lab
- Cool design makes the difference, lovely homes
- Long term value is short term value
- Without passion, you cannot do it!

Session 92:

Addressing the challenges and creating opportunities for sustainable development in the Mediterranean region: What role can the UfM Secretariat play?

Chairperson:

Hoballah, Arab

Chief Sustainable Consumption and Production of UNEP/DTIE (Division of Technology, Industry and Economics)

Speakers:

Zalloom, Bushra
Birmingham City University, Jordania.
Sherif, Yasser
Egyptian Forum for Sustainable development, Egipto.
Fleuret, Guy
Senior Advisor, Transport & Urban Development Division.
Guerber, François

Senior Advisor, Energy and Climate Change Division.

Conclusions:

This is the chair summary (not approved by the audience)

1. UfM, a co-owned initiative/partnership, focusing on flagship projects in various sectors, but not on policies- through an important financing mechanism. UPFI to support projects that are innovative and mature.
2. Med. region being increasingly urbanized, cities will be the drivers of sustainability in the region, with focus on Resource Efficiency, aiming also at reducing pollution of the Mediterranean Sea.
3. Transformative approach should consult with and involve concerned stakeholders if to promote liveability and sustainability. Affordability need to be included in urban planning: need to humanize cities by socializing policies.
4. Huge needs in the Mediterranean Region, therefore rational for various initiatives, but value added from and through UfM yet not very clear. Need to better show and communicate about the rationale of UfM and the added value it brings to the busy landscape of North-South Mediterranean cooperation.

Session 93:

Are current building regulations adequately advancing sustainable buildings? If not, what is missing, and how should they be changed? (I)

Chairperson:

Dodds, Bill

Scottish Government. Scotland

Speakers:

Title: Impact of energy efficiency goals on systems of building regulations and control
Visscher, Henk
Delft University of Technology. Delft.Netherlands.
Title: Perspective of the Building Sustainability Regulatory Evolution in Spain: From Prescription to Performance
Serra, Javier
Ministry of Infrastructure. Navarra.Spain.
Title: Reviewing challenges between the need for government-subsidised housing in South Africa and the sustainability requirements of the National Building Regulations
Laubscher,Jacques
Tshwane University of Technology. South Africa.
Title: Adapting Building Regulatory Systems to Better Address Climate Change Impacts

Meacham, Brian J.

Worcester Polytechnic Institute. Worcester, MA.USA.

Conclusions:

- A broader adaptable framework for building regulation is needed.
- We need to look at social an economic capacity
- We need to be aware of improvement to existing buildings so that they do not have unintended consequences.
- Safe, healthy sustainable development
- There is a need for retrospective change to existing buildings
- Look at risk assessed approach to solutions for tackling differing objectives for existing buildings.

Session 94:

Shall comfort and well-being be actually the top arguments to develop and sell sustainable buildings?

Chairperson:

Szpunar, Slawomir

International Marketing Director, Saint-Gobain

Speakers:

Folch, Marc
Architect, Calderon-Folch-Sarsanedas Arquitectes.
Sanders, Helen
VP Technical business development, Sage Glass.
Le Bars, Marie-Annick
Innovation Director, Bouygues Immobilier.
Blyth, Alastair

Design Strategist: Learning Environments.

Conclusions:

Yes. Comfort and wellbeing are strongest arguments to promote sustainable buildings. When numerous arguments of productivity gains exist for non residential buildings (schools, offices, hospitals), we are still struggling to find convincing evidence (beyond health) for residential sector, programs like Saint-Gobain multi-comfort are an efficient way to demonstrate relevance and benefits of sustainable approach for comfort and wellbeing.

Session 96:

How can urban planning, infrastructure and mobility management improve sustainable development in our cities?

Chairperson:

Villalante, Manel

Industrial Engineer Associate Director Mobility International Projects BARCELONA REGIONAL Urban Development Agency

Speakers:

Garriga, Victoria
Architect. Barcelona.Spain.
Shetty, Prassad
Chief Planning Division City of Mumbai. Mumbai.India.
Guallart, Vicente

Chief Architect City of Barcelona. Barcelona.Spain.

Conclusions:

La ciudad de Bagdad (Victoria Garriga) ha presentado el plan estratégico de la ciudad en el horizonte 2030 destacando aspectos de infraestructuras y movilidad, contextualizando Bagdad en su actual situación de conflicto social, político y bélico.
La ciudad de Mumbai (Prashad Shetty) ha presentado una reflexión sobre la realidad social y formalidad de esta ciudad fuertemente densa y con problemas de superpoblación.
La ciudad de Barcelona ha reflexionado sobre sus actuales políticas y estrategias en movilidad a infraestructuras para mejorar la ciudad y a sus ciudadanos.
Ha disertado sobre la combinación entre los diferentes medios de transporte (intermodalidad).
<u>Conclusiones</u>
Tres ciudades muy diferentes, tres visiones muy diferentes pero con puntos de coincidencia:

- Necesidad de apostar por el transporte público y la reducción del uso del vehículo privado.
- Apostar por la ciudad compacta y policéntrica
- Avanzan en el funcionamiento de un sistema de movilidad integrado (intermodalidad) y en red
- No existen fórmulas mágicas y las políticas deben de adaptarse a la morfología de cada ciudad y a su realidad social, económica y cultural.

Session 98:

Knowledge transfer between the business and universities worlds?

Chairperson:

Casaldáliga Albisu, Pau

CEIM Coordinator

Speakers:

Sin Yu Lau, Stephen
Hong Kong University (HKU). Hong Kong.China.
Guivernau Molina, Jordi
Technical department, ROCKWOOL PENINSULAR S.A.U. Barcelona.Spain.
Ruiz del Portal, Angela
Research assistant on the Welsh Energy Sector Training Project (WEST). CARDIFF University.
González Canto, Bejamín

Corporate Development Director, CYPE Ingenieros, S.A.

Conclusions:

Sponsoring is required to go forward with any new and innovative university initiative. Frequency, degree of similarity, loss of knowledge after transfer, knowledge about the own organization, fundamental ITEM from this transfer.

Session 110:

Which are the keys to dealing with the technological changes that the building renovation sector requires?

Chairperson:

Paricio, Ignacio

ITEC (Instituto de Tecnología de la Construcción de Catalunya). Architect and Professor in Universidad de Arquitectura de Barcelona. Barcelona, Spain

Speakers:

Solé, Josep
European Sustainability and Technical Manager at URSA Insulation. Barcelona.Spain.
Ripoll, Angel
Marketing at Technal. Barcelona.Spain.
Saiz Sánchez, Pablo
Architect, founding partner at Modulab architecture and housing. Madrid.Spain.
Saurina, Daniel

Ingeniero. Director comercial de Disset. Barcelona.Spain.

Conclusions:

Rehabilitation is quite complex, and could be expensive. We need to develop, architects, technicians and industry, prefabricated ad hoc, flexible and polyfunctional elements to rehabilitate.
Big companies, architects, contractors and technicians must work together in order to save materials, money. We must work together to optimize and make rehabilitation easy and affordable.

Session 111:

We are moving. But are EU-Member States moving in the same direction?

Chairperson:

Hamans, Chris

European Sustainability Consulting, Heel, The Netherlands

Speakers:

Asensão, G.
CEN-CENELEC - European Standardisation Organisation,. Brussels.Belgium.
Trudsø, E.R.
Nordic Innovation. Oslo.Norway.
Brockmann, T.
Head of Division - Federal Institute for Research on Building, Urban Affairs and Spatial Development. Berlin.Germany.
De Lathauwer, D.

Federal Public Service Health; DG5; Environment and Product Policy. Brussels.Belgium.

Conclusions:

No we are not moving in the same direction! The intentions are there but no real **common** effort. We should prioritise and focus on core information (not too scientific) and make it user friendly to apply. Simplification but not limitation.
There are barriers that need a roadmap and capacity.
Common principles are widely accepted by all member states (LCA, EN 15804) but specific elements (e.g. background data, databases) need to be lined up.

Session 112:

Which are the limits and how can we achieve nearly zero emission office buildings today?

Chairperson:

Irastorza Ruigomez, Luis

Tecnopen

Speakers:

Álvarez Domínguez, Servando
Universidad de Sevilla. Sevilla.Spain.
Pascual Pellicer, Jordi
AIGUASOL . Sistemes Avançats d'Energia Solar Tèrmica, SCCL. Spain.
Expósito Mora, Carlos
ALIA, Arquitectura, Energía y Medio Ambiente, S.L.
Ingelmo Torres, Alfredo

Gas Natural Distribución, SDG, S.A. Spain.

Conclusions:

- A divorce of the two pillars on which sustainable buildings have been based during the last few years is clearly taking place due to the large amount of resources found both for oil and gas.
- Therefore, the need for sustainable buildings is based mainly on the climate change issue and on the economic return of the investment in energy efficiency.
- The EU has very strict objectives for 2030 with the 30% reduction of GHG emissions relative to 1990, the 27% of renewables share in the final energy and the 27% increase in energy.
- The building envelopes need dramatic changes in the future if highly energy efficient buildings are required.
- The construction costs of very efficient buildings is not bigger than for conventional buildings.
- Very small difference between the points of minimum cost and minimum energy.
- Relevance of passive design and real operation conditions.
- Double glazing buildings are always very far of the minimum global cost of the building (investment + operations & maintenance – residual value)

Session 113:

Responsible business practice in land, construction and real estate

Chairperson:

Hartenberger, Ursula

Global Head of Sustainability, RICS

Speakers:

Title: Promoting responsible business practice in land, construction and real estate
Pes Guixa, Angel
Brand and Corporate Responsability Director, Caixa Bank, President, Global Compact Network Spain.
Title: Human Rights, Labour, Environment and Anti-Corruption: Mapping sectoral engagement with the Ten Principles of the UN Global Compact
Cosgrove, Alexander
Associate, Latham & Watkins.
Title: Best practice example II: Embedding environmental best practice in construction
Antink, Roy
Director Green Support, Skanska.
Title: Best practice example III: Building capacity for corporate sustainablity

El Burai, Mahmoud Hesham

Director, Dubai Real Estate Institute.

Conclusions:

- Need to strengthen stakeholder engagement and communication.
- Widen the sustainability agenda to account for social issues more.

Session 114:

What are the current health & social inclusive drivers for the sustainable development of Cities?

Chairperson:

Wassouf, Micheel

Director. Energiehaus scp

Speakers:

Hellweg, Uli
IBA-Hamburg, Germany. Hamburg.Germany.
Ng, John
Director, Hong Kong Green Building Council, Hong Kong Special Administrative Region. Hong Kong.
Orsini, Francesco

Urban Eafit, Medellin, Colombia. Medellín.Colombia.

Conclusions:

There is no general answer to this question:

- Reducing emissions and improving live condition
- Ensuring social and cultural diversity
- Decentralised approach
- Covering top-down with bottom-up strategies

1. Labelling is a good social driver for transformation towards liveability and sustainability
2. Laboratories as motors driving new urban planning strategies
3. Foster education to integrate inequity

Municipality is a facilitator: Social equity can only be established by environmental equity, with an involved mayor.

Session 115:

Is the Energy Performance Certificate (EPC) of the Energy Performance of Buildings Directive (EPBD) promoting building renovation?

Chairperson:

Morer Forns, Lluís

Head of Energy Efficiency. Catalan institute of Energy (ICAEN)

Speakers:

Domínguez Martín, Aitor
Head of Domestic & Buildings / Technical Project. Ministry of Industry, Energy and Tourism Spain. Spain.
Thers Egesberg, Leise
Legal advisor Danish Energy Agency.
Garcia Audí, Pau
Project Officer Executive Agency for Small and Medium Enterprises (EASME). Unit B1 – Energy Efficiency.
López, Fabián
Societat Orgànica, Consultora ambiental S.L.

Conclusions:

- The EPC is useful in different in different fields
- The EPC is the starting point for public policies
- To consider EPC as a valid economic and technical tool, it is absolutely necessary to increase reliability = quality
- EPC just give information, but it could be a powerful tool to promote renovation

Session 116:

Do the SB Conferences really serve their purpose?

Chairperson:

Miguel Mitre, Emilio

GBCe and WSB14 Barcelona, SB13 Coordinator and WSB14 Barcelona International Representative

Speakers:

Title: What have we talked about at the SB Conferences? A thematic structure as an analytical, adaptive a tool to know the contents of the SB Conference Series as a whole, and to be able to plan its future evolution. Methodology.
Miguel Mitre, Emilio
GBCe and WSB14 Barcelona, SB13 Coordinator and WSB14 Barcelona International Representative.
Title: What's the History and planned Evolution of the SB Conference Series?
Larsson, Nils
Representative of the four co-promoting organizations, and Overall Series Coordinator, iiSBE (international initiative for a Sustainable Built Environment).
Title: Is the SB Conference Series a good way for the research world to actively contribute to Sustainable Building?
Bakens, Wim
CIB (International Council for Research and Innovation in Building and Construction).
Title: What is the relevance of the work done by iiSBE at the SB Conference Series?
Bragança, Luis
iiSBE President / Professor of Sustainable Building at the Civil Engineering Department of University of Minho.
Title: How can UNEP-SBCI approach be more present at the SB Conferences?
Garrigan, Curt
UNEP-SBCI (Sustainable Building and Climate Initiative of the United Nations' Environmental Programme).
Title: Are the SB Conferences an adequate field for the pragmatic, technical approach of the world's largest engineers federation?
Baillon, François
FIDIC (International Federation of Consulting Engineers).

Conclusions:

They are very good at certain points:

- They enable people to meet and are a source of inspiration

They could be improved in various ways:

- More concentrated on performance
- Wider global coverage
- Becoming more urban
- Joining forces with other organizations
- Having more participation of the Regional SB Conferences, and getting a much

stronger feedback from the Regional Conferences

- Strongly supporting developing countries' attendance
- Integrating research and practice
- Accepting contributions aside the traditional peer review process, and/or by improving this process, and making it also possible at other, differentiated levels
- With more in depth integration of policy makers and economy views
- Improved dissemination, starting with an after –Conference newsletter with all the conclusions
- Seeking economic additional support to be able to do the prior. The industry could provide this

We have the knowledge but we need to progress improving our sensitivity

Session 127:

How can BIM help designing sustainable buildings?

Chairperson:

Jallon, Paul

Director of Development at Knauf, France

Speakers:

Häkkinen, Tarja
Doctor Sc Tech. Senior principal research scientist at VTT. Technical Research Centre of Finland. Finland.
Coloma, Eloi
BIM Innovation Specialist. PhD Lecturer at University UPC.. Barcelona.Spain.
Mills, Fred
Procurement Manager at Osborne and Founding Director of The B1M (BIM one million). United Kingdom.

Conclusions:

Of course BIM helps us design new sustainable buildings, by:

- A process around the tool itself, which gives time to answer good questions (energy savings, comfort, acoustics, safety)
- Sharing data with IFC format for example, of the products linked to the building design. Data that could be used for management of the buildings, maintenance, deconstruction at the end of life.
- BIM has a great potential and may be at the beginning. It should be sustainable itself and it seems to be to make buildings sustainable.

Session 128:

Which are the real barriers to renovation?

Chairperson:

Martínez, Juan Carlos

Journalist specialiced in real state sector, cooperates with El País, Inmodiario, Economía Digital

Speakers:

Mateo, Luis
Director Andimat, Spanish National Association of Insulation Materials Manufacturers. Madrid.Spain.
Gómez, Carlos
CEO NVR, New Vision on Refurbishment. Madrid.Spain.
Etchenique, Ana
Vice-chairman CECU, Spanish Confederation of Consumers and Users. Madrid, Spain.
González, Penélope
Representative of a Condominium renovated with high standard insulation. Madrid, Spain.

Conclusions:

There are two main barriers: political and social. Politicians are not interested in energy efficiency, neither in renovation. There is a lack of interest for them. It doesn't happen with the automotive sector, they push politicians and they obtain public investment.
On the other hand, people are not interested enough, we need an educational change. People have to push politicians more. Right now, it is possible to save money and to improve comfort, but the government and banks don't make it easy.

Session 131:

Putting sustainability in real estate into practice

Chairperson:

Bosteels, Tatiana

Head of Responsible Property Investment, Hermes Real Estate, UNEP FI Property Working Group Co-Chair

Speakers:

Title: Strengthening the business case for energy efficiency in buildings in Europe: The role of the European Commission
Garcia, Pau
Executive Agency for Small and Medium-sized Enterprises, European Commission.
Title: Sustainability metrics in real estate investment and management
Hovorka, Frank
MRICS, Head of Sustainable Development, Caisse des Dépôts, UNEP FI Property Working Group Co-Chair.
Title: Greening the building supply chain
Garrigan, Curt
Coordinator UNEP SBCI.
Title: RenoValue: State of play regarding the integration of sustainability in valuation practice in Europe
Hartenberger, Ursula
Global Head of Sustainability, RICS.
Title: Sustainability from a valuation practioner´s perspective
Creamer, Mark
MRICS, Head of EMEA Valuation & Advisory Services CBRE.

Conclusions:

There are a lot of thoughts:

- About dissemination: need to translate the data/knowledge/tools
- Right level of information addressed, to right level of decision making
- Need to up-skill wide range of real estate and construction practitioners

Session 132:

Is it possible to achieve a friendly and affordable urban district retrofitting?

Chairperson:

Moro, Andrea

iiSBE Italia R&D

Speakers:

Essig, Natalie
MUAS - Munich University of Applied Sciences - Germany. Germany.
Ferrando, Valeria
IES - Integrated Environmental Solutions Limited. United Kingdom.
Kiss, Ida
ABUD - Hungary.
Camiruaga, Idoia
City of Santiago - Spain. Santiago.Spain.

Conclusions:

Yes, but only if support to decision tool is available to support urban retrofitting processes. Tools should connect the stakeholders in a very efficient process. It is necessary to have, beside the tools, an integrated process to optimize solutions.
Integrated processes supported by tools to take the right decisions. Process + tool. Need of tools for decision making. Communication with user and political will is fundamental. monitoring+ education.

Session 137:

Can Active House be the tool that drives sustainable buildings to become standard solutions for buildings?

Chairperson:

Eriksen, Kurt Emil

Active House Alliance, Bruxelles, Belgium

Speakers:

Zsolt Gunther, Zsolt
3h architecture. Budapest.Hungary.
Valentin, Emmanuel
Saint Gobain. Paris.France.
Schjodt Worm, Amdi
Danish Technological Institute. Aarhus.Denmark.
Christiansen, Niels B.
Danish Technical University. Copenhagen.Denmark.

Conclusions:

Yes, it is an easy understandable tool that can be a compass to show directions and to be used as a dialog tool.
You should focus on different levels for different building types, as well as new build and renovation.
Evaluate how to use data from other schemes and use the tool for monitoring also.
(Agreed by audience)

Session 138:

Institutionalizing Sustainability: Transforming Professional Practices to Achieve Portfolio-wide improvements / consistently Higher Performance on Projects

Chairperson:

Batshalom, Barbra

Executive Director, Sustainable Performance Institute

Speakers:

Ekman, Rand
Director of Sustainability, Cannon Design.
McEvoy, Paula
Co-director Sustainable Design, Associate Principal, Perkins+Will.
Hubbard, Gunnar
Principal and Building Sustainability Practice Leader, Thornton Tomasetti.

Conclusions:

Conclusions:

1. Important to engage people through culture: events, speakers, "earth day", internal seminars.
2. Research and developments (leads to innovation) are critical to advance best practices. Large firms can afford it, small firms can partner with universities.
3. Frameworks are key to help achieve success. AIA 2030 challenge that makes you measure and track. SPI, new firm that gives roadmaps to institutional sustainable and John Kotter's Change Management Framework.

Session 144:

SB Challenge. Do our buildings perform as intended?(II)

Chairperson:

Rietz, Andreas

Speakers:

Müller, Julia
Germany.
Loehnert, Günter
Germany.
Possemiers, Toon
Bruxelles.Belgium.
Atkins, Richard
United Kingdom.
Larsson, Nils
iiSBE. Canada.

Conclusions:

- The buildings presented have demonstrated to be both best practice as new or generally modernized buildings and operated buildings in terms of sustainability and energy efficiency. However, it is obvious that even these buildings need a certain time until they will operate (nearly) the same way as intended.
- Sustainable Building Certification should be applied as a proper instrument for optimization all over the project phases in order to meet the original project targets required and authorized by the client.
- While the monitoring will provide information on the real performance of a building, the optimization of building operation is necessary to tune the building performance as intended.